Toxicology & Clinical Pharmacology Consensus

Serotonin Syndrome: Clinical Diagnosis, Washout Protocols & Triage

Reference Code: CPG-SEROTONIN-2026.03 | **Target Audience:** Psychiatrists, Emergency Physicians, Clinical Pharmacists

1. Pathophysiology & Clinical Triad

Serotonin Syndrome (serotonin toxicity) is a life-threatening, drug-induced toxidrome caused by excessive agonism at central and peripheral serotonergic receptors (specifically 5-HT1A and 5-HT2A). The condition is characterized by a classical triad of: **(1) Neuromuscular Hyperactivity** (clonus, hyperreflexia, myoclonus, rigidity); **(2) Autonomic Hyperactivity** (diaphoresis, hyperthermia, tachycardia, tachypnea, labile BP); and **(3) Mental Status Alterations** (agitation, delirium, confusion, coma). Unlike Neuroleptic Malignant Syndrome (NMS), serotonin toxicity develops acutely within hours of drug ingestion or dose escalation.

2. Hunter Toxicity Criteria (Gold Standard Diagnostic Framework)

In the presence of a known serotonergic agent, Serotonin Syndrome is definitively diagnosed if ANY of the following rules are met:

Rule #	Mandatory Clinical Sign	Associated Symptoms Required	Diagnostic Sensitivity / Specificity
Criterion 1	Spontaneous Clonus	None required (sufficient on its own)	Sensitivity 84%, Specificity 97%
Criterion 2	Inducible Clonus	PLUS Agitation OR Diaphoresis	High diagnostic specificity
Criterion 3	Ocular Clonus (slow horizontal)	PLUS Agitation OR Diaphoresis	Early sensitive neurological indicator
Criterion 4	Tremor	PLUS Hyperreflexia (especially in lower extremities)	Common presentation in mild-to-moderate toxicity
Criterion 5	Hypertonia	PLUS Core Temperature > 38.0°C (100.4°F) PLUS Ocular/Inducible Clonus	Indicator of severe, immediately life-threatening toxicity

3. High-Risk Drug Combinations & Absolute Contraindications

Drug Class 1	Drug Class 2	Mechanism of Synergy	Clinical Severity & Mandated Action
MAO Inhibitors (Phenelzine, Tranylcypromine, Linezolid, Methylene Blue)	SSRIs / SNRIs (Sertraline, Fluoxetine, Venlafaxine)	Simultaneous blockade of serotonin degradation (MAO-A) and reuptake inhibition causes massive, catastrophic synaptic 5-HT surge.	**ABSOLUTELY CONTRAINDICATED. High risk of fatal hyperthermia (>41°C), rhabdomyolysis, DIC, and death. Strict washout required.**
SSRIs / SNRIs (Sertraline, Venlafaxine)	Tramadol	Tramadol is both a μ-opioid agonist and a serotonin/norepinephrine reuptake inhibitor (SNRI). Causes severe additive toxicity.	**AVOID COMBINATION**. Substitute with non-serotonergic analgesics (e.g., Oxycodone, Morphine, Acetaminophen).
SSRIs / SNRIs / Tricyclics	Triptans (Sumatriptan, Zolmitriptan)	Triptans act as selective 5-HT1B/1D agonists; co-administration with reuptake inhibitors enhances peripheral & central receptor activation.	FDA Black Box Advisory. Inform patients of clonus/agitation warning signs. Limit triptan frequency.
SSRIs / SNRIs	Lithium	Lithium enhances presynaptic release and postsynaptic sensitivity of central serotonergic neurons.	Use lowest possible doses. Monitor for tremor, ataxia, confusion, and clonus during any titration.
Sertraline / Fluoxetine	Bupropion	Bupropion is a potent CYP2D6 inhibitor, raising SSRI/SNRI plasma concentrations up to 4-fold while lowering seizure threshold.	Monitor antidepressant levels; reduce SSRI dose by 50% when initiating bupropion.

4. Mandatory Antidepressant Washout Periods

Switching between serotonergic antidepressants requires strict adherence to pharmacokinetic washout intervals calculated based on 5 elimination half-lives (t½) of the parent drug and active metabolites:

Initial Agent Discontinued	Elimination Half-Life	Mandatory Washout Before MAOI	Clinical Rationale
Fluoxetine (Prozac)	Parent: 2–4 days Active norfluoxetine: **7–15 days**	**5 WEEKS (35 Days)**	Norfluoxetine metabolite persists for over a month. Premature MAOI initiation has resulted in fatalities.
Sertraline (Zoloft)	Parent: 26 hours Active N-desmethylsertraline: 66 hrs	**2 WEEKS (14 Days)**	Safe clearance achieved after 14 days.
Venlafaxine (Effexor)	Parent: 5 hours O-desmethylvenlafaxine: 11 hrs	**7 to 14 Days**	Short half-life, but severe discontinuation syndrome requires gradual taper.
MAOIs (Phenelzine)	Irreversible enzyme inactivation	**2 WEEKS (14 Days)**	Requires 14 days for regeneration of functional endogenous MAO enzymes before starting SSRI.

5. Emergency Triage & Management Protocol

Severity Grade	Presenting Signs	Pharmacologic Interventions	Level of Care
Mild	Hyperreflexia, mild tremor, diaphoresis, normal temp, alert	Discontinue all serotonergic agents. Oral Diazepam 5-10 mg for restlessness. Reassurance and observation.	Outpatient observation or 24-hr Clinical Decision Unit (CDU).
Moderate	Inducible clonus, tachycardia, mydriasis, temp 38.0–38.9°C, agitation	IV fluids, IV Lorazepam 2-4 mg. **Cyproheptadine: 12 mg oral initial loading dose**, then 2 mg every 2 hours if symptoms persist (max 32 mg/day).	Telemetry or Step-Down Inpatient Unit.
Severe (Life Threatening)	Spontaneous clonus, rigidity, temp > 39.0°C (102.2°F), delirium, metabolic acidosis	Immediate RSI endotracheal intubation. Neuromuscular paralysis with Vecuronium (avoid succinylcholine). Aggressive active external cooling.	Intensive Care Unit (ICU) admission.